THE CURIOSITY CUP 2025

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Predictive Model Trends: Predicting Early Detection of Alzheimer's Disease using SAS Viya Workbench for Learner®

Team Data ACES

ABSTRACT

This report presents the correlation found for predicting early detection of Alzheimer's disease using the SAS Viya Workbend for Learner®. It also explores three datasets of different periods *(7 months ago, 1 year ago, and 5 years ago)* to identify differences in their approach to prediction. These datasets were found in Kaggle and used for the analyses: *Alzheimer's Disease Prediction (*Tarek, A., *2024)*, *EDA | Classification | AlzheimerDisease (*zj, 2023), and *detecting-early-alzheimer-s-hyperparameter-0-86 (*Ramirez, R., 2019). The aim was to find the dataset with the highest accuracy in predicting Alzheimer's disease and the correlation between the variables and the disease. By doing so, better datasets can be developed by identifying and including significant variables that strongly influence the prediction of Alzheimer's disease

INTRODUCTION

Alzheimer's disease, one of the leading causes of dementia, is a progressive neurodegenerative disorder that affects and destroys brain cells, leading to memory loss, cognitive dysfunction, and sudden behavioral changes. According to National Health Service (2018), there is currently no cure for Alzheimer's disease. As the disease progresses, it can impair cognitive and motor functions, leading to dehydration, poor nutrition, and increased susceptibility to infections, ultimately leading to death. Given these, early detection and treatment are viable in slowing disease progression and improving patient's general well-being.

This paper aims to address and develop a predictive model for the early detection and onset of Alzheimer's disease and identify patterns and correlations between different parameters associated with Alzheimer's disease. Additionally, by analyzing multiple datasets from different periods, we aim to improve early prediction of the disease onset. This research could contribute to earlier interventions, improved disease management, and ultimately, a reduction in the burden of Alzheimer's on individuals, families, and healthcare systems.

DATA PREPARATION

Three datasets were obtained from Kaggle, which was sourced from *Alzheimer's Disease Prediction (*Tarek, A., *2024)*, *EDA | Classification | AlzheimerDisease (*zj, 2023), and *detecting-early-alzheimer-s-hyperparameter-0-86 (*Ramirez, R., 2019). The authors cleaned the data on the datasets by removing unnecessary columns, dropping rows with null values and duplicated values, and turning categorical data into numerical data, resulting in the following records and features: (2149, 35), (212, 8), (373, 15). The datasets

alzheimers_disease_data.csv, Plasma lipidomics in Alzheimers disease and its progression-1.csv, and oasis_longitudinal.csv were downloaded and transformed into a SAS data file by using PROC IMPORT. This transformed data file was then uploaded into SAS Viya Workbench for Learner to use in the study.

PROBLEM STATEMENT

This study aims to develop a predictive model for the early detection of Alzheimer's disease by analyzing different parameters and by comparing three distinct periods of datasets *(7 months ago, 1 year ago, and 5 years ago)*. This research seeks to identify patterns that improve prediction accuracy, allowing for earlier interventions and better disease management, ultimately reducing the global impact of Alzheimer's.

DATA CLEANING AND VALIDATION

DATASET 1: EDA | Classification | AlzheimerDisease

On the first imported dataset, the feature 'Diagnostic' was designated as the target variable–it had three labels: Alzheimer's Disease, Mild Cognitive Impairment, and Control. The dataset contained missing values in CSF Amyloid, CSF Total tau, CSF Phosphorylated tau, and APOE4, which were imputed using listwise deletion. Since no duplicates were found, no records were dropped. The following features were dropped due to irrelevance: Sample, Progression to Alzheimer's Disease, and Progression time (months). To ensure compatibility with machine learning models, categorical variables were converted into numerical representations. The Diagnostic column was mapped so that Alzheimer's Disease was represented as 1, while Mild Cognitive Impairment and Control were both set to 0, effectively grouping non-Alzheimer's conditions into a single category. The Sex column was transformed where Male was encoded as 0 and Female as 1. Similarly, the APOE4 column was mapped to 1 for Yes and 0 for No, ensuring a standardized numerical format for genetic risk factors. Afterward, a feature correlation heatmap was constructed, revealing that MMSE (-0.53), CSF Amyloid (-0.33), CSF Total tau (0.29), and Age (0.32) were highly correlated with Alzheimer's diagnosis.

Figure 1.1 presents the feature importance scores obtained from a Gradient Boosting model trained on Dataset 1 to predict Alzheimer's disease. The chart illustrates the relative contribution of seven features to the model's predictive power. MMSE (Mini-Mental State Examination) demonstrates the highest degree of importance, significantly outweighing all other features. CSF Amyloid levels and Age also exhibit substantial importance scores. In contrast, APOE4 status and Sex show comparatively minimal influence on the model's predictions. This visualization highlights the prominent role of cognitive assessment (MMSE) and biomarker analysis (CSF Amyloid), in conjunction with age, in the predictive framework developed from Dataset 1.

DATASET 2: Alzheimer's Disease Prediction

On the second dataset, the feature 'Diagnosis' was set as the target variable–the diagnosis status for Alzheimer's Disease, where 0 indicates No and 1 indicates Yes. The variables 'PatientID' and 'DoctorInCharge' were dropped. A correlation heatmap revealed that Memory Complaints (0.31) and Behavioral Problems (0.22) show a positive correlation with Alzheimer's diagnosis. This suggests that self-reported memory issues and behavioral changes are early signs of cognitive decline. On the other hand, MMSE (-0.24), ADL (-0.33), and Functional Assessment (-0.36) have negative correlations, meaning lower scores on these assessments indicate a higher likelihood of Alzheimer's. This aligns with clinical expectations—patients with Alzheimer's struggle with daily living tasks and cognitive tests.

Figure 1.2 displays the feature importance scores derived from a Gradient Boosting model trained on Dataset 2 for the prediction of Alzheimer's disease. The visualization presents the relative influence of 16 clinical and demographic variables on the model's predictive capacity. Functional Assessment and MMSE (Mini-Mental State Examination) scores emerged as the most influential predictors, exhibiting substantially higher importance scores compared to other features. Age also demonstrated a notable degree of importance. Conversely, factors such as cardiovascular disease, cholesterol levels (LDL and triglycerides), depression, and head injury exhibited comparatively low levels of importance in this model. This suggests that functional and cognitive assessments, along with age, are the primary drivers of the model's predictive performance on Dataset 2.

DATASET 3: MRI and Alzheimers

On the third dataset, the feature 'Group' was designated as the target variable–the dataset was composed of three categories: Demented, Nondemented, and Converted. Attributes such as 'Subject ID', 'MRI ID', and 'Hand' were dropped. Then, handling missing values in SES and MMSE were dropped accordingly. Since no duplicates were found, no records were dropped. Then, categorical data were converted into numerical format. The target variable 'Group' mapped 1 for Demented while 0 for Nondemented and Converted. A correlation heatmap revealed that CDR (0.81) showed the strongest positive correlation with Alzheimer's diagnosis, indicating that higher dementia ratings are highly associated with disease onset. SES (0.21) had a mild positive correlation, suggesting that socioeconomic factors might slightly influence disease progression. On the other hand, EDUC (-0.23) and M/F (-0.26) had negative correlations, meaning that higher education levels and gender differences might contribute to variations in Alzheimer's susceptibility, with higher education potentially serving as a protective factor. Similarly, nWBV (-0.29) exhibited a negative correlation, reinforcing that lower brain volume is a key indicator of Alzheimer's disease. MMSE (-0.62) had the strongest negative correlation, indicating that lower cognitive function scores are highly predictive of Alzheimer's onset.

Figure 1.3 illustrates the feature importance scores generated by a Gradient Boosting model trained on Dataset 3 for Alzheimer's disease prediction. The chart displays the relative importance of ten features in contributing to the model's predictive capability. Notably, CDR (Clinical Dementia Rating) exhibits a remarkably high importance score, significantly surpassing all other features. MMSE (Mini-Mental State Examination) and MR Delay show

moderate levels of importance. Conversely, features such as ASF (Atlas Scaling Factor), Age, EDUC (Education), M/F (Gender), SES (Socioeconomic Status), Visit (Visit Number), eTIV (Estimated Total Intracranial Volume), and nWBV (Normalized Whole Brain Volume) demonstrate relatively low importance scores. This visualization emphasizes the dominant role of the Clinical Dementia Rating in the predictive model derived from Dataset 3.

DATA ANALYSIS

For predictive modeling, the Gradient Boosting Classifier was implemented as the primary predictive model. Model performance was evaluated using Accuracy Score, Confusion Matrix, F1-Score, and ROC-AUC Curve, and hyperparameters were fine-tuned using GridSearchCV. The findings indicated that the Gradient Boosting Model was the most effective, achieving high predictive accuracy.

DATASET 1: EDA | Classification | AlzheimerDisease

The best hyperparameters identified for Dataset 1 were learning_rate = 0.2, max_depth = 3, min_samples_split = 5, n_estimators = 10, and subsample = 0.8. The model achieved a best cross-validation accuracy of 0.7436 and a test accuracy of 0.7500.

Considering *Figure 2.1*, the classification report for Dataset 1 showed an overall accuracy of 75%, with class 0 (non-Alzheimer's) achieving a precision of 0.74 and recall of 0.81, while class 1 (Alzheimer's) had a precision of 0.76 and recall of 0.68. The macro-average and weighted F1-score were both 0.75, indicating a balanced performance across both classes.

Figure 2.2 depicts the training loss (deviance) of a Gradient Boosting model trained on Dataset 1. The loss is plotted against the number of boosting iterations, which range from 1 to 10. The chart shows a consistent decrease in training loss with each iteration, indicating the model's improvement in predictive accuracy during the training process.

DATASET 2: Alzheimer's Disease Prediction

For Dataset 2, the best hyperparameters identified were learning_rate = 0.2, max_depth = 2, min_samples_split = 2, n_estimators = 25, and subsample = 0.8. The model achieved a best cross-validation accuracy of 0.9529 and a test accuracy of 0.9558.

In line with *Figure 3.1*, the classification report for Dataset 2 showed an overall accuracy of 95%, with class 0 (non-Alzheimer's) achieving a precision of 0.95 and recall of 0.97, while class 1 (Alzheimer's) had a precision of 0.95 and recall of 0.92. The macro-average F1-score was 0.95, indicating a high-performing and well-balanced model.

Figure 3.2 provided illustrates the training loss (deviance) of a Gradient Boosting model trained on Dataset 2. The loss is plotted against the number of boosting iterations, ranging from 0 to 25. The chart demonstrates a rapid decrease in training loss during the initial iterations, followed by a gradual decline, suggesting effective learning and convergence of the model.

DATASET 3: MRI and Alzheimers

For Dataset 3, the best hyperparameters identified were learning_rate = 0.2, max_depth = 5, min_samples_split = 5, n_estimators = 150, and subsample = 1.0. The model achieved a best cross-validation accuracy of 0.9577 and a test accuracy of 0.9577.

Figure 4.1 showed an overall accuracy of 94%, with class 0 (non-Alzheimer's) achieving a precision of 0.96 and recall of 0.96, while class 1 (Alzheimer's) had a precision of 0.92 and recall of 0.92. The macro-average and weighted F1-score were both 0.94, indicating a strong and well-balanced model for classifying Alzheimer's and non-Alzheimer's cases.

Figure 4.2 displayed the training loss (deviance) of a Gradient Boosting model trained on Dataset 3. The loss is plotted against the number of boosting iterations, spanning from 0 to 100. The chart reveals a fluctuating yet overall downward trend in training loss as iterations increase, suggesting a gradual improvement in the model's predictive performance during training.

CONCLUSION

Based on *Table 1*, this study successfully developed a predictive model for the early detection of Alzheimer's disease using Gradient Boosting Classifiers on three distinct datasets. The models demonstrated varying degrees of accuracy.

In conclusion, the analysis reinforces the potential of machine learning in aiding early Alzheimer's detection. The high accuracy achieved by the models suggests that these tools could be valuable assets in clinical settings, assisting healthcare professionals in identifying individuals at risk and facilitating timely interventions. Identifying significant variables in early detection is paramount in Alzheimer's management, as it allows for the implementation of strategies to slow disease progression and improve patient quality of life.

SUGGESTIONS FOR FUTURE STUDIES

Future Alzheimer's disease prediction studies can be improved by utilizing the significant variables found in the paper. Variables such as CDR (Clinical Dementia Rating), MMSE (Mini-Mental State Examination), Functional Assessment, CSF Amyloid levels, and Age can be given priority for Alzheimer's disease prediction as they've shown the highest correlation with the disease. Furthermore, other machine learning models such as decision trees, random forest, and SVM can be used to identify the best model to use for the prediction.

REFERENCES

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National Health Service (NHS). (2018). Alzheimer's disease overview. https://www.nhs.uk

World Health Organization (WHO). (2023). Dementia. https://www.who.int

APPENDIX

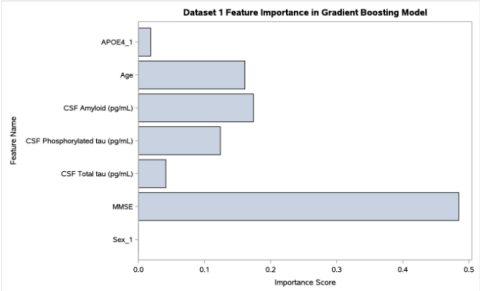

Figure 1.1 EDA | Classification | AlzheimerDisease Dataset - Feature Importance

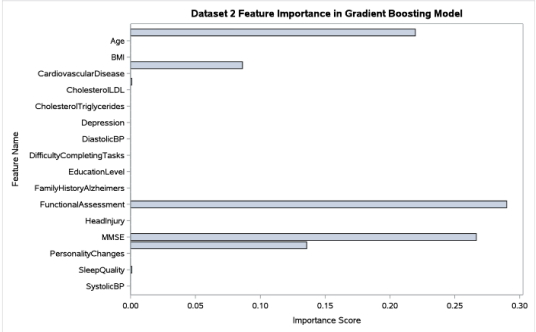

Figure 1.2 Alzheimer's Disease Prediction Dataset - Feature Importance.

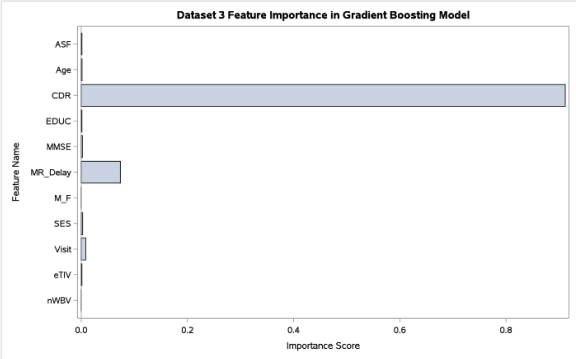

Figure 1.3 MRI and Alzheimer's Dataset - Feature Importance

Dataset 1 Plasma Lipidomics Data Classification Report (Best Model)

Class Label	Precision	Recall	F1-Score	Support
0	0.7400	0.8100	0.7700	21
1	0.7600	0.6800	0.7200	19
Accuracy	.	.	0.7500	40
MacroAvg	0.7500	0.7500	0.7500	40
Weighted	0.7500	0.7500	0.7500	40

Figure 2.1 EDA | Classification | AlzheimerDisease Dataset - Classification Report

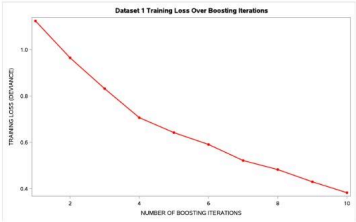

Figure 2.2 EDA | Classification | AlzheimerDisease Dataset - Training Loss

Dataset 2 Alzheimer's Disease Data Classification Report (Best Model)

Class Label	Precision	Recall	F1-Score	Support
0	0.9500	0.9700	0.9600	277
1	0.9500	0.9200	0.9300	153
Accuracy	.	.	0.9500	430
MacroAvg	0.9500	0.9400	0.9500	430
Weighted	0.9500	0.9500	0.9500	430

Figure 3.1 Alzheimer's Disease Prediction Dataset - Classification Report

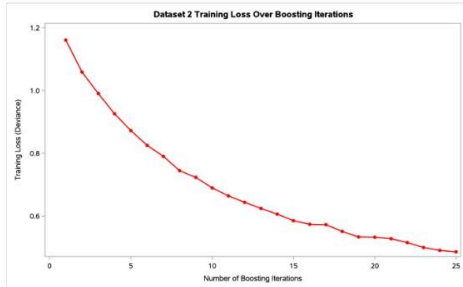

Figure 3.2 Alzheimer's Disease Prediction Dataset - Training Loss

Dataset 3 OASIS Longitudinal Data Classification Report (Best Model)

Class Label	Precision	Recall	F1-Score	Support
0	0.9600	0.9600	0.9600	45
1	0.9200	0.9200	0.9200	26
Accuracy	.	.	0.9400	71
MacroAvg	0.9400	0.9400	0.9400	71
Weighted	0.9400	0.9400	0.9400	71

Figure 4.1 MRI and Alzheimer's Dataset - Classification Report

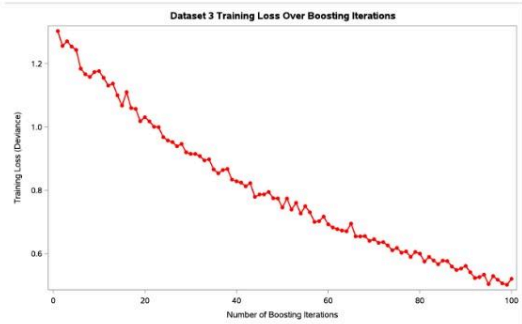

Figure 4.2 MRI and Alzheimer's Dataset - Training Loss

Dataset	Accuracy	Number of Iteration	Top 3 Significant Variables
EDA | Classification | AlzheimerDisease	75%	10	MMSE, CSF Amyloid levels, and Age
Alzheimer's Disease Prediction	95%	25	Functional Assessment, MMSE, and Age
MRI and Alzheimer's	94%	100	CDR, MMSE, and MR Delay

Table 1. Comparison of Gradient Boosting Accuracy and Significant Variables